FRAME CONSTRUCTION SPECIFICATION

01 Navigator and 02 Butterfly. Titanium, in eight colourways.

SUMMARY

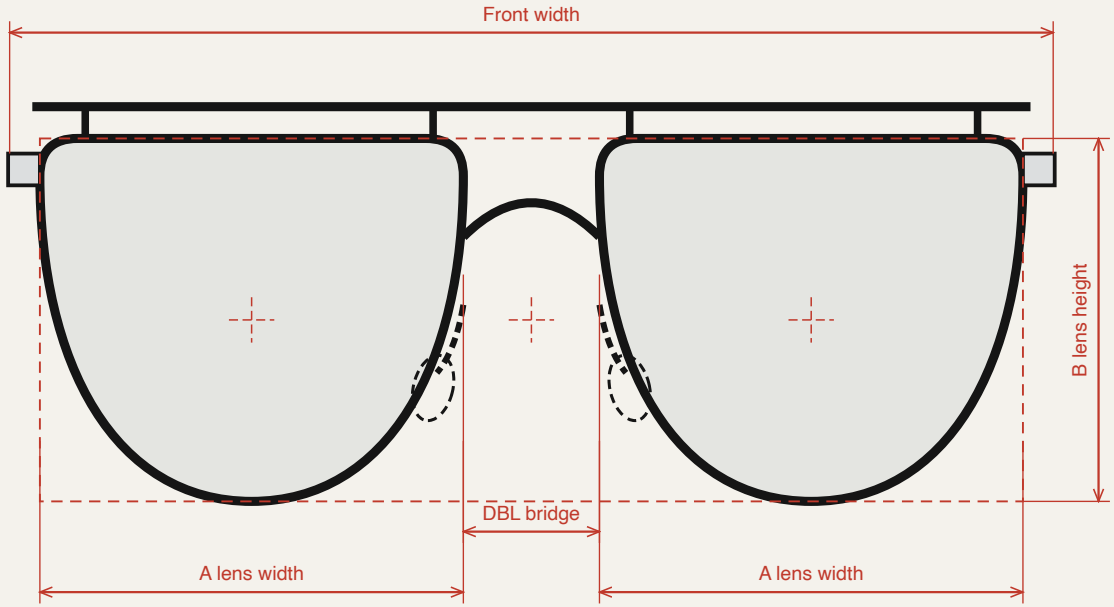
Two frames drawn in one fine line of titanium wire. This document fixes every dimension, part, material, finish and tolerance needed to model, prototype and manufacture them.

ITEM	01 NAVIGATOR	02 BUTTERFLY
Wearer	Men	Women
Shape	Straight brow bar over a rounded lower lens	Wide butterfly, outer corners lifted 4 mm
Lens width A	56 mm	54 mm
Lens height B	48 mm	46 mm
Bridge DBL	18 mm	18 mm
Temple length	140 mm	140 mm
Front width	138 mm	135 mm
Box code	56 □ 18 140	54 □ 18 140
Colourways	Fleur, Gris, Noir, Ambre	Fleur, Rose, Ambre, Noir

COMMON TO BOTH	VALUE
Material	Titanium wire and sheet
Rim wire	1.2 mm diameter
Temple	1.6 mm wire, flattened to 3.6 x 1.7 mm for 50 mm from the hinge
Hinge	Two barrel, 1.2 mm screw
Nose pads	Adjustable titanium arm, clear silicone pad 11 x 7 mm
Lenses	Polarised, UV400, 2.0 mm centre thickness
Face form angle	5 degrees
Pantoscopic angle	8 degrees
Engraving	VXIII and the first line of the verse, inner face of each temple

MEASURING SYSTEM

Measurements follow the boxing system of ISO 8624. Each lens is enclosed in a box, A is the width of the box, B is its height and DBL is the distance between the two boxes. A frame is marked A □ DBL, then the temple length.

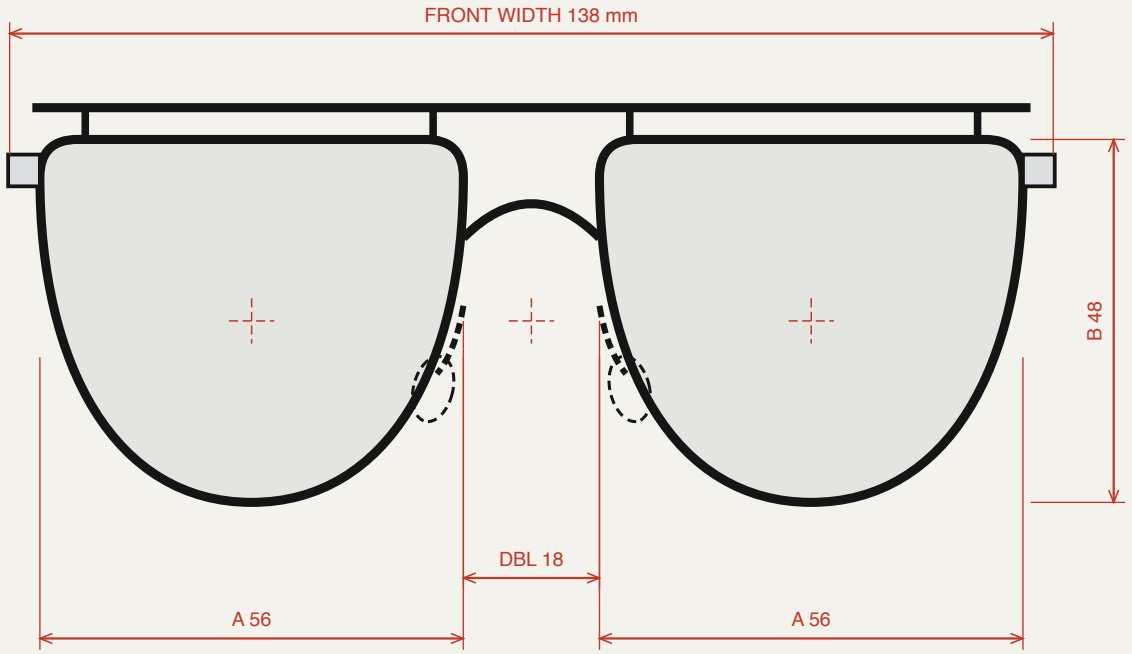


Boxing system, ISO 8624. The box encloses each lens and DBL is measured between the boxes.

TERM	DEFINITION
A	Horizontal width of the lens box.
B	Vertical height of the lens box.
DBL	Distance between the lenses, measured between the boxes.
Front width	Overall width across the outer faces of the end pieces.
Temple length	From the hinge axis to the tip, along the wire.
Length to bend	From the hinge axis to the point where the temple drops.
Face form angle	Angle of each half of the front to the frontal plane.
Pantoscopic angle	Forward tilt of the front from vertical.

ALL DIMENSIONS ARE IN MILLIMETRES. DRAWINGS ARE AT 1:1 UNLESS A SCALE IS GIVEN.

01 NAVIGATOR, FRONT VIEW

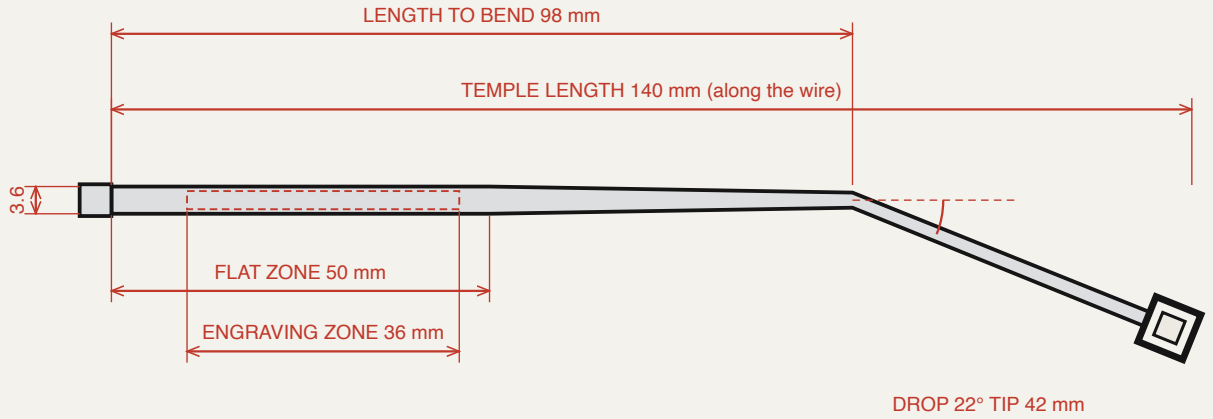


SCALE 1:1
Lens centre distance 74 mm. Dashed lines show the pad arms behind the lens.

RIM
Titanium wire 1.2 mm, grooved 0.5 mm to take the lens bevel.

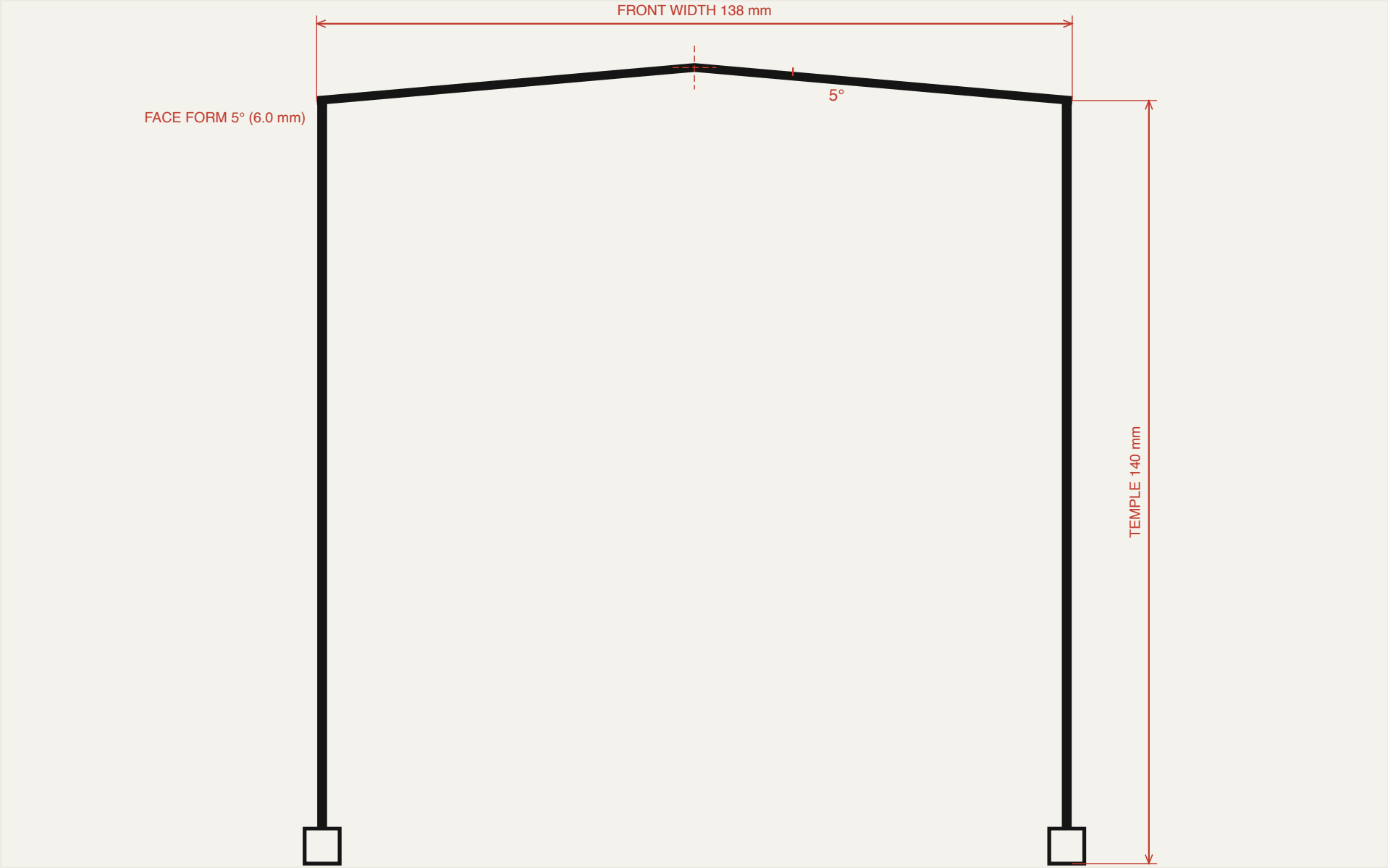
BROW BAR
Straight, 1.2 mm wire, 4.2 mm above the lens top, joined by two posts each side.

01 NAVIGATOR, TEMPLE, SIDE VIEW



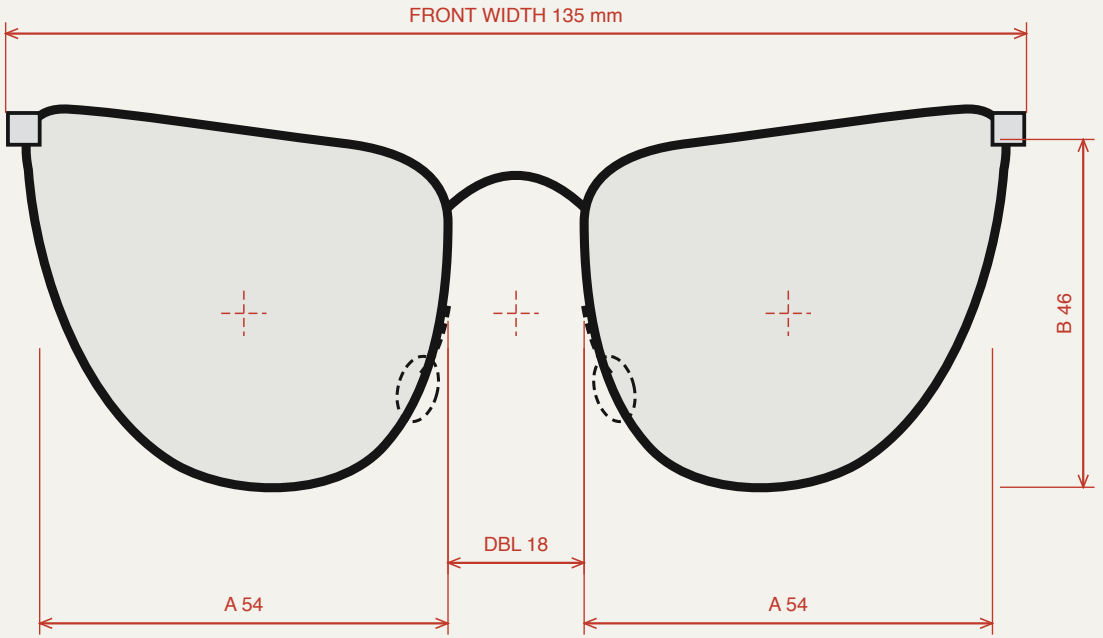
ZONE	FROM HINGE	SECTION	PURPOSE
Flat zone	0 to 50 mm	3.6 x 1.7 mm	Carries the engraving on the inner face
Taper	50 to 98 mm	2.4 x 1.6 mm falling to 2.0 mm	Spring and comfort
Drop	98 to 133.8 mm	Round, 1.6 mm	Sits behind the ear at 22 degrees
Tip	133.8 to 140 mm	Open square 6.4 mm	The cube, drawn as a single line

01 NAVIGATOR, PLAN VIEW



OVERALL OPEN WIDTH 138 mm at the end pieces

02 BUTTERFLY, FRONT VIEW

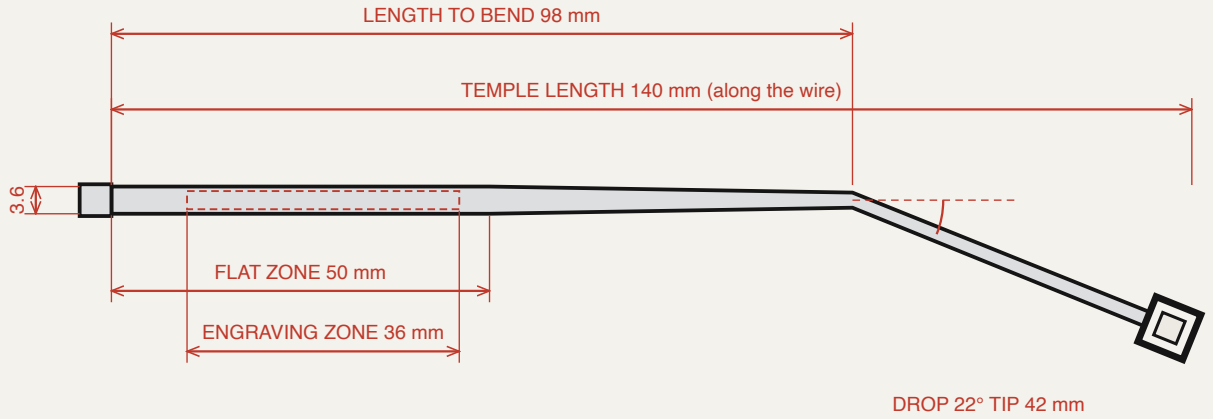


SCALE 1:1
Lens centre distance 72 mm. Dashed lines show the pad arms behind the lens.

RIM
Titanium wire 1.2 mm, grooved 0.5 mm to take the lens bevel.

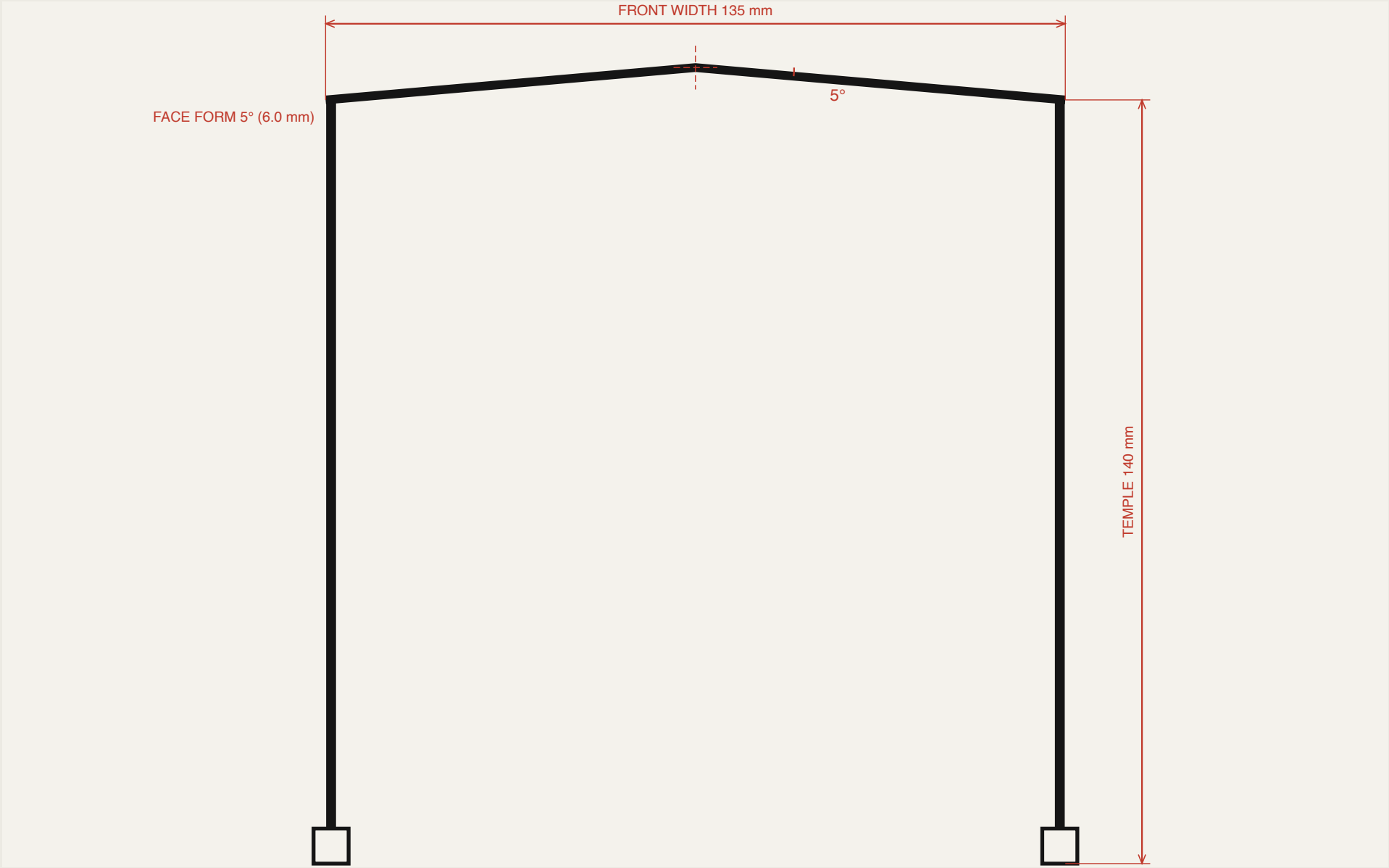
BRIDGE
Single arched wire, 1.2 mm, rising 8 mm at its centre.

02 BUTTERFLY, TEMPLE, SIDE VIEW



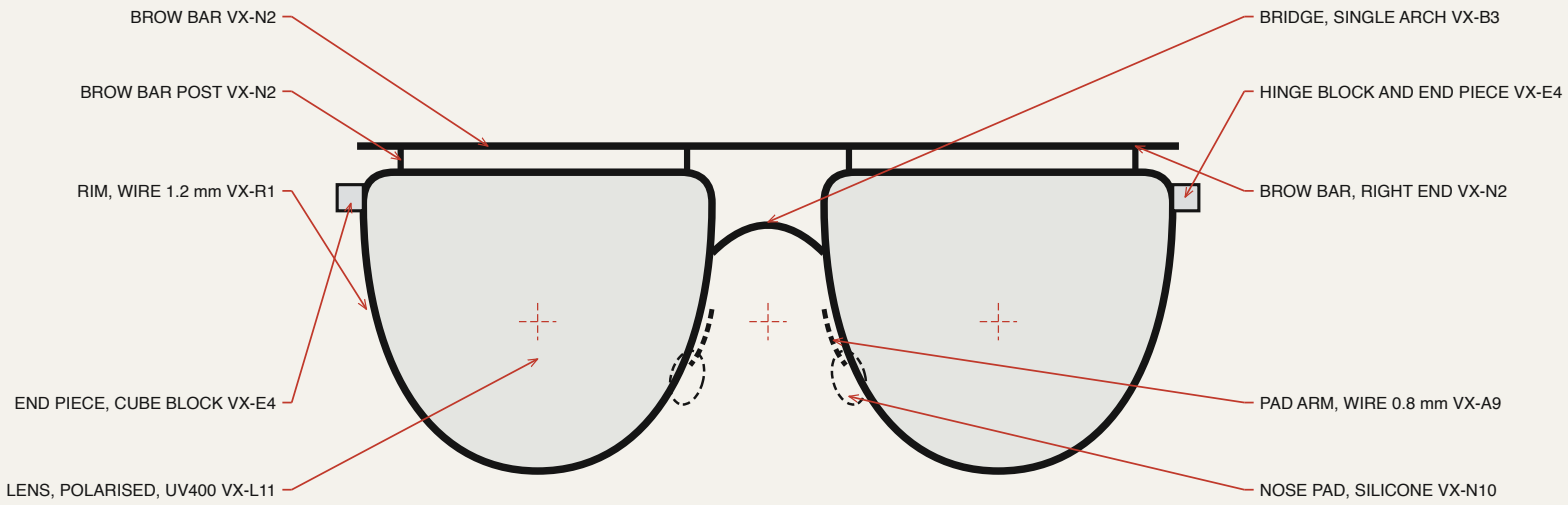
ZONE	FROM HINGE	SECTION	PURPOSE
Flat zone	0 to 50 mm	3.6 x 1.7 mm	Carries the engraving on the inner face
Taper	50 to 98 mm	2.4 x 1.6 mm falling to 2.0 mm	Spring and comfort
Drop	98 to 133.8 mm	Round, 1.6 mm	Sits behind the ear at 22 degrees
Tip	133.8 to 140 mm	Open square 6.4 mm	The cube, drawn as a single line

02 BUTTERFLY, PLAN VIEW



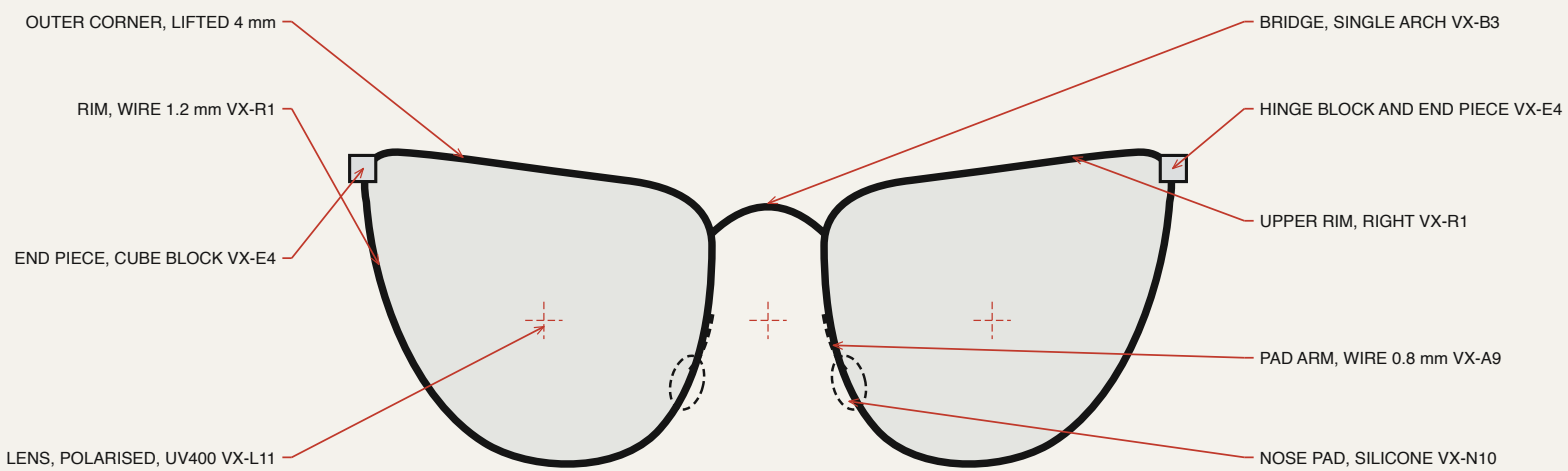
OVERALL OPEN WIDTH 135 mm at the end pieces

PARTS LABELLED, 01 NAVIGATOR, FRONT



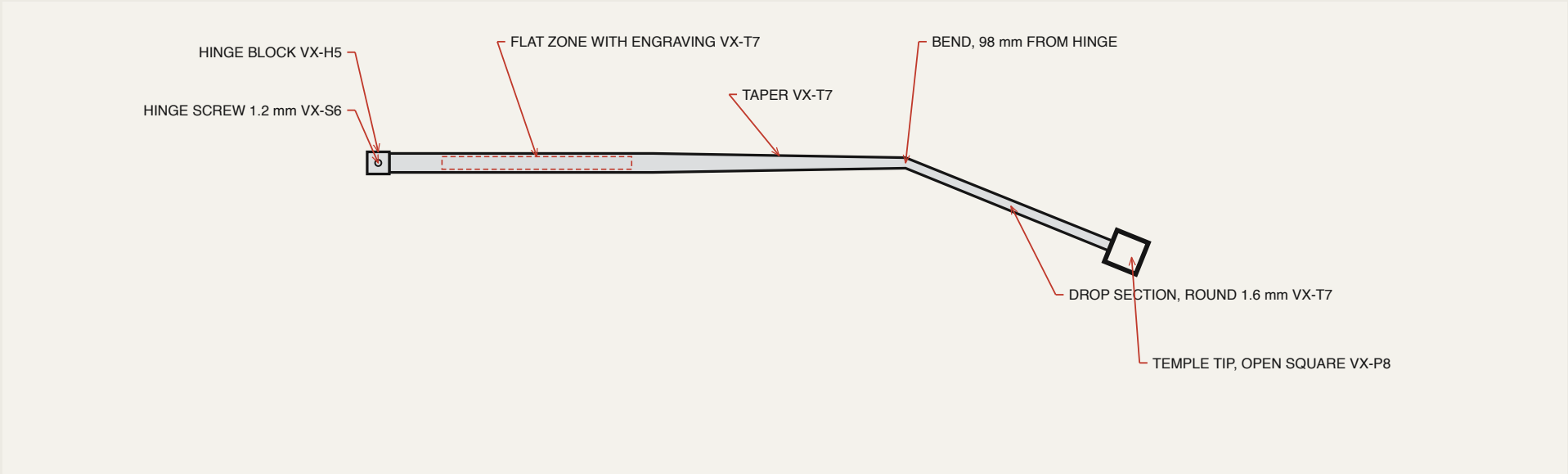
EACH ARROW POINTS TO THE PART NAMED AT ITS TAIL. PART NUMBERS MATCH THE PARTS AND MATERIALS TABLE.

PARTS LABELLED, 02 BUTTERFLY, FRONT



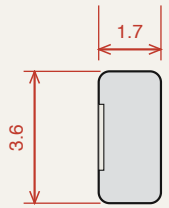
EACH ARROW POINTS TO THE PART NAMED AT ITS TAIL. THE BUTTERFLY HAS NO BROW BAR AND ITS UPPER RIM CARRIES THE LIFTED CORNER.

PARTS LABELLED, TEMPLE AND MARKING



TOP: TEMPLE IN SIDE VIEW. BOTTOM: INNER FACE OF THE FLAT ZONE OF EACH TEMPLE AT ABOUT 3:1, SHOWING THE ENGRAVED MARKS.

TEMPLE SECTIONS AND ENGRAVING ZONE



SECTION A, FLAT ZONE

3.6 x 1.7 mm, engraved 0.14 mm



SECTION B, TAPER

2.4 x 1.6 mm at 75 mm



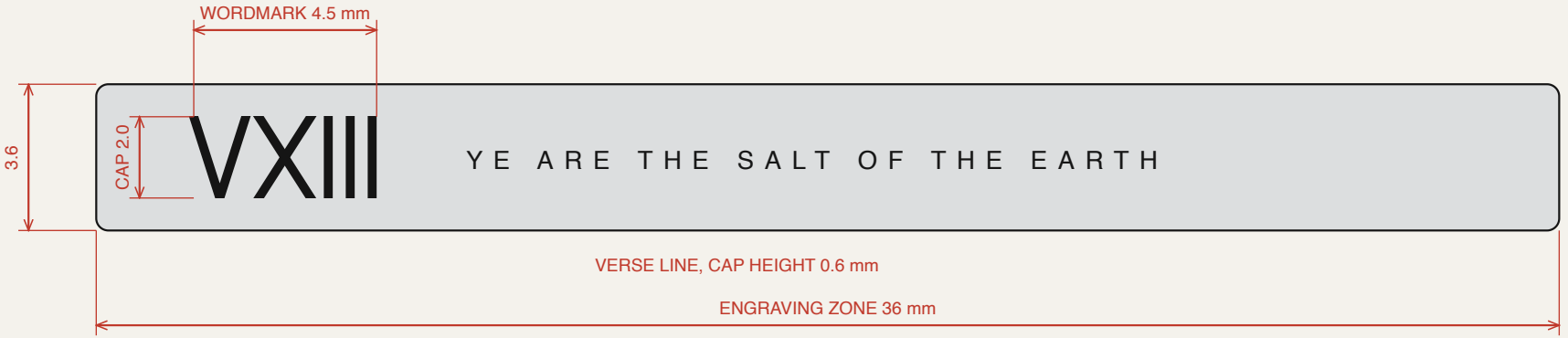
SECTION C, ROUND WIRE

Diameter 1.6 mm beyond the bend



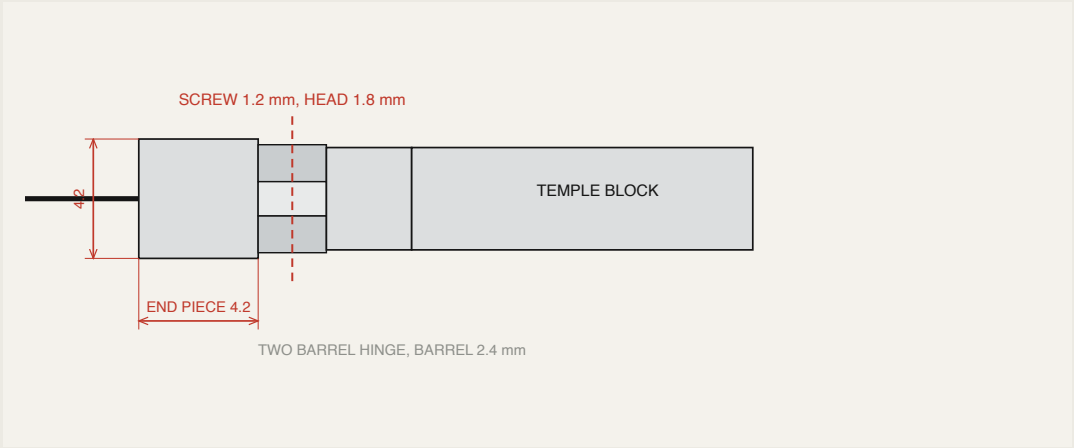
SECTION D, RIM WIRE

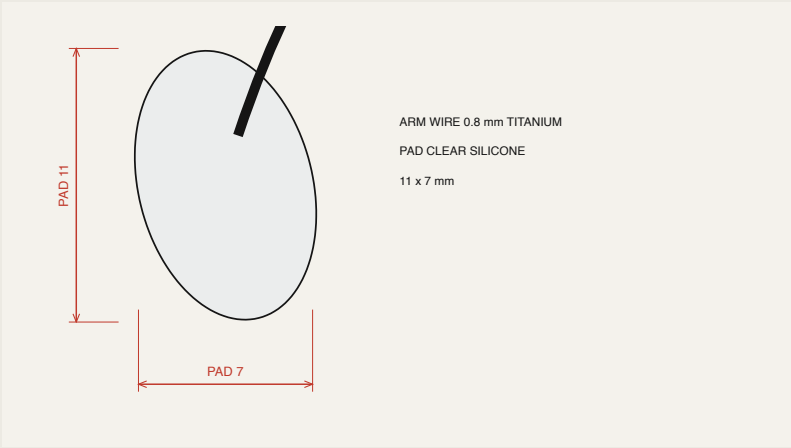
Diameter 1.2 mm, lens groove 0.5 mm



SECTIONS AT 5:1. ENGRAVING ZONE AT 6:1. THE WORDMARK IS CUT 0.14 MM DEEP AT A CAP HEIGHT OF 2.0 MM, THE ZONE BEGINS 10 MM FROM THE HINGE AXIS AND THE WORDMARK STARTS 2.4 MM INSIDE IT.

HINGE, END PIECE AND NOSE PAD

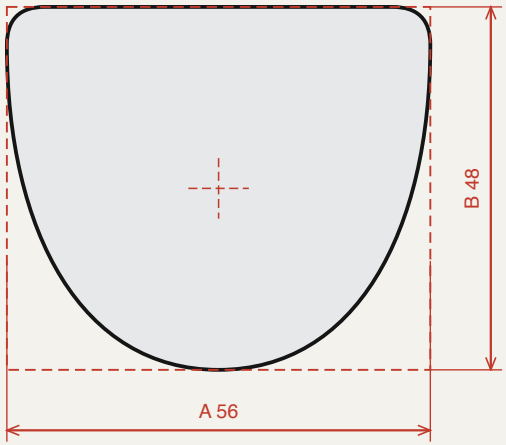




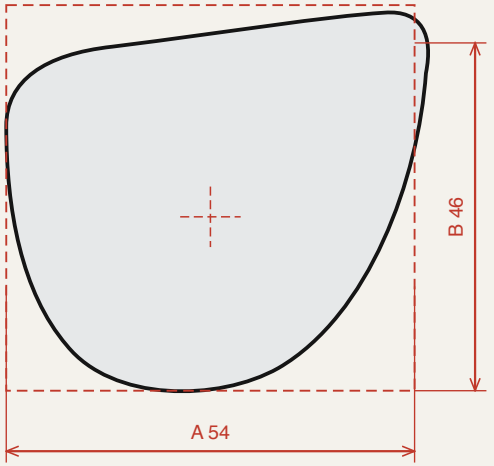
PART	SPECIFICATION
End piece	Titanium block 4.2 mm cube with square edges, laser welded to the rim wire.
Hinge	Two barrel hinge with 2.4 mm barrels, 1.2 mm titanium screw, head 1.8 mm, locking compound on the thread.
Nose pad arm	0.8 mm titanium wire, adjustable, welded at the lens nasal edge.
Nose pad	Clear silicone, 11 x 7 mm, push fit onto the pad arm.

LENS TEMPLATES

01 NAVIGATOR LENS, RIGHT



02 BUTTERFLY LENS, RIGHT



ITEM	SPECIFICATION
Material	Polarised lens, single tint per frame as set in the colourway table
Protection	UV400, blocking ultraviolet light up to 400 nm
Centre thickness	2.0 mm
Edge	V bevel to fit the 0.5 mm rim groove
Coating	Hard coat and oleophobic outer coat
Marking	None on the lens. The brand is carried only by the engraving.

FIT

PARAMETER	01 NAVIGATOR	02 BUTTERFLY	NOTE
Front width	138 mm	135 mm	Suits a face width of about 135 to 145 mm.
Lens centre distance	74 mm	72 mm	Close to the common adult pupillary distance.
Temple length	140 mm	140 mm	With a drop of 22 degrees at 98 mm.
Face form angle	5 degrees		Sets the front slightly around the face.
Pantoscopic angle	8 degrees		Tilts the lower lens edge toward the cheek.
Vertex distance	12 mm		From the back of the lens to the cornea.
Nose pad spacing	17 mm at the lower edge	16 mm at the lower edge	Adjustable through the pad arms.
Weight	Target 18 g	Target 16 g	Without case.

PARTS AND MATERIALS

PART	NUMBER	QUANTITY	MATERIAL	FINISH
Rim, left and right	VX-R1	2	Titanium wire 1.2 mm	As colourway
Brow bar and posts (Navigator)	VX-N2	1 set	Titanium wire 1.2 mm and 1.0 mm	As colourway
Bridge	VX-B3	1	Beta titanium wire 1.2 mm	As colourway
End piece	VX-E4	2	Titanium block 4.2 mm	As colourway
Hinge	VX-H5	2 pairs	Titanium, two barrel	As colourway
Hinge screw	VX-S6	2	Titanium 1.2 mm	As colourway
Temple	VX-T7	2	Titanium wire 1.6 mm, flattened	As colourway, flat zone brushed
Temple tip	VX-P8	2	Titanium, open square 6.4 mm	As colourway
Pad arm	VX-A9	2	Beta titanium wire 0.8 mm	As colourway
Nose pad	VX-N10	2	Clear silicone	None
Lens	VX-L11	2	Polarised, UV400	Hard coat

FINISH AND LENS BY COLOURWAY

Every visible metal part takes the colourway finish, including the screws and the temple tips. The nose pads stay clear and the lens carries the single tint shown.

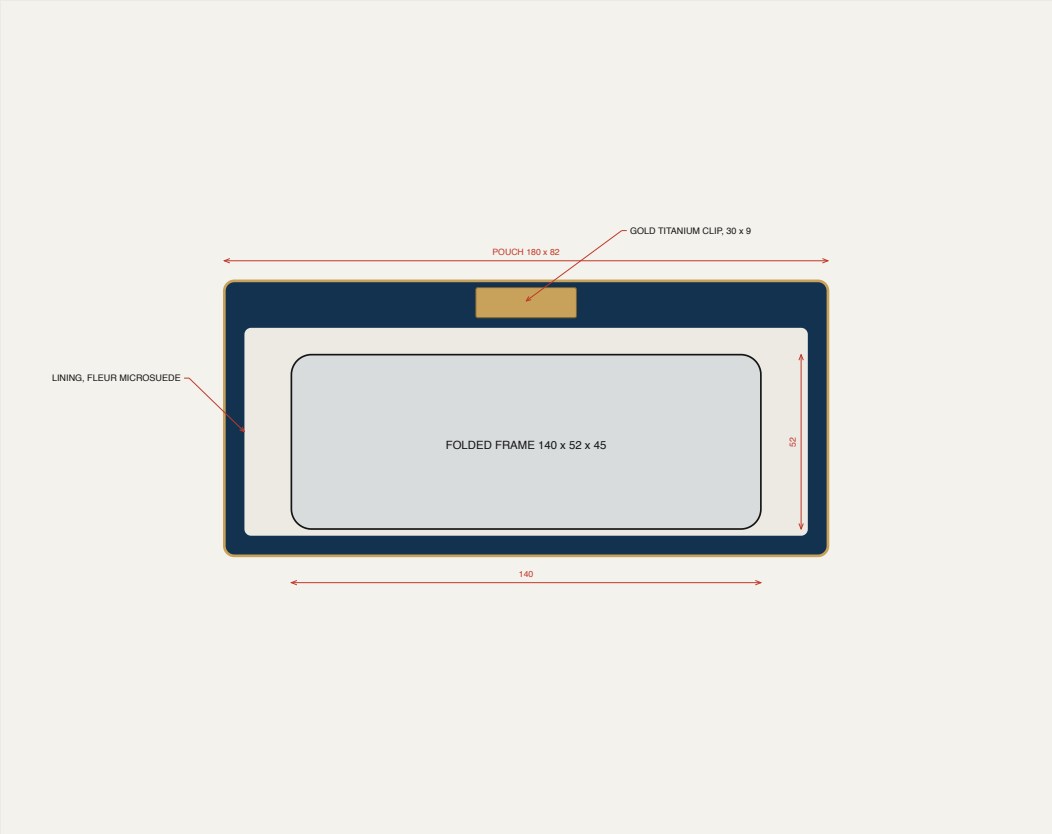
NAME	METAL FINISH	LENS	PROCESS
Fleur	Polished silver titanium #C9CCCE	Clear grey #A9ACAB	Polish, then clear anodise
Gris	Brushed gunmetal titanium #4A4D50	Smoke #4B4540	Brush, then dark PVD
Noir	Matte black titanium #151515	Deep grey #2A2B2C	Bead blast, then black PVD
Rose	Rose gold titanium #C9988A	Soft amber #D9A66B	Polish, then rose PVD
Ambre	Warm gold titanium #C8A25A	Amber #C9812A	Polish, then gold PVD

RANGE	FRAMES
01 Navigator	01 Fleur, 01 Gris, 01 Noir, 01 Ambre
02 Butterfly	02 Fleur, 02 Rose, 02 Ambre, 02 Noir

ENGRAVING AND MARKING

ITEM	SPECIFICATION
Brand mark	VXIII, in the identity wordmark, one word, cap height 2.0 mm, line 0.22 mm, cut 0.14 mm deep.
Verse line	YE ARE THE SALT OF THE EARTH, in capitals, cap height 0.6 mm, set to the right of the wordmark with a gap of 2.2 mm.
Position	Inner face of the flat zone on both temples, the engraving zone beginning 10 mm from the hinge axis and the wordmark starting 2.4 mm inside it.
Method	Laser engraving through the finished colour, after the finish is applied, so the cut shows the bare titanium beneath.
Size mark	Inside the left temple, beyond the verse line: 56 □ 18 140 or 54 □ 18 140, cap height 0.8 mm.
Product mark	Inside the right temple, beyond the verse line: the product name, for example 01 FLEUR.
Regulatory	Conformity mark on the inner left temple where the market requires it, and UV400 on the packaging.
Packaging match	The box label and the certificate of authenticity repeat the product name and the number marked inside the right temple.
Not marked	Lens. Front of the frame. Outer face of the temple.

PACKAGING FIT



ITEM	SPECIFICATION
Folded frame	140 × 52 × 45 mm with the temples folded behind the lenses
Pouch	180 × 82 mm, lined microsuede, interior about 170 × 70 mm
Clearance	At least 12 mm at each end and 9 mm top and bottom
Weight in the box	Frame about 18 g, with the box about 400 g
Frame in pouch	Lenses face the lining and the nose pads rest against the leather

THE BOX, ITS LAYERS, THE NOTES AND THE POUCH ARE SET OUT IN THE PACKAGING DOCUMENT.

MANUFACTURE

STEP	OPERATION	CONTROL
01	Form wire for rim, bridge and pad arms	Template check at 1:1
02	Flatten and taper temple wire	Section check at 20, 50 and 75 mm
03	Laser weld rims, bridge, end pieces and posts	Weld visual and pull test
04	Polish or brush, then clean	Surface against the master sample
05	Apply anodise or PVD colour	Colour against the master, thickness record
06	Engrave the inner face of the temples through the finish	Depth 0.14 mm, zone from 10 mm
07	Assemble hinge, screw and pads	Torque and swing test
08	Cut and fit lenses	Lens fit without stress, A and B within tolerance
09	Final inspection and packing	Dimensional check, cosmetic check, UV test per batch

TOLERANCES AND TESTING

DIMENSION	TOLERANCE
Lens width A	± 0.2 mm
Lens height B	± 0.3 mm
Bridge DBL	± 0.3 mm
Temple length	± 1.0 mm
Front width	± 0.5 mm
Rim wire diameter	± 0.03 mm
Engraving depth	± 0.03 mm
Face form and pantoscopic angle	± 1 degree

TEST	REQUIREMENT
Sunglass lens requirements	Conforms to ISO 12312-1 for the filter category chosen
Spectacle frame requirements	Conforms to ISO 12870 for dimensional and robustness tests
Corrosion	Artificial perspiration test, no visible change to finish
Hinge cycle	No loosening after the stated opening cycles
Nickel release	Within the limit of the market sold into
Finish adhesion	No flaking after the tape and abrasion test

THE MASTER SAMPLE OF EACH COLOURWAY IS THE REFERENCE FOR COLOUR AND SURFACE AND IS KEPT SEALED.